SpecialistRoute Handover — 1. Guide

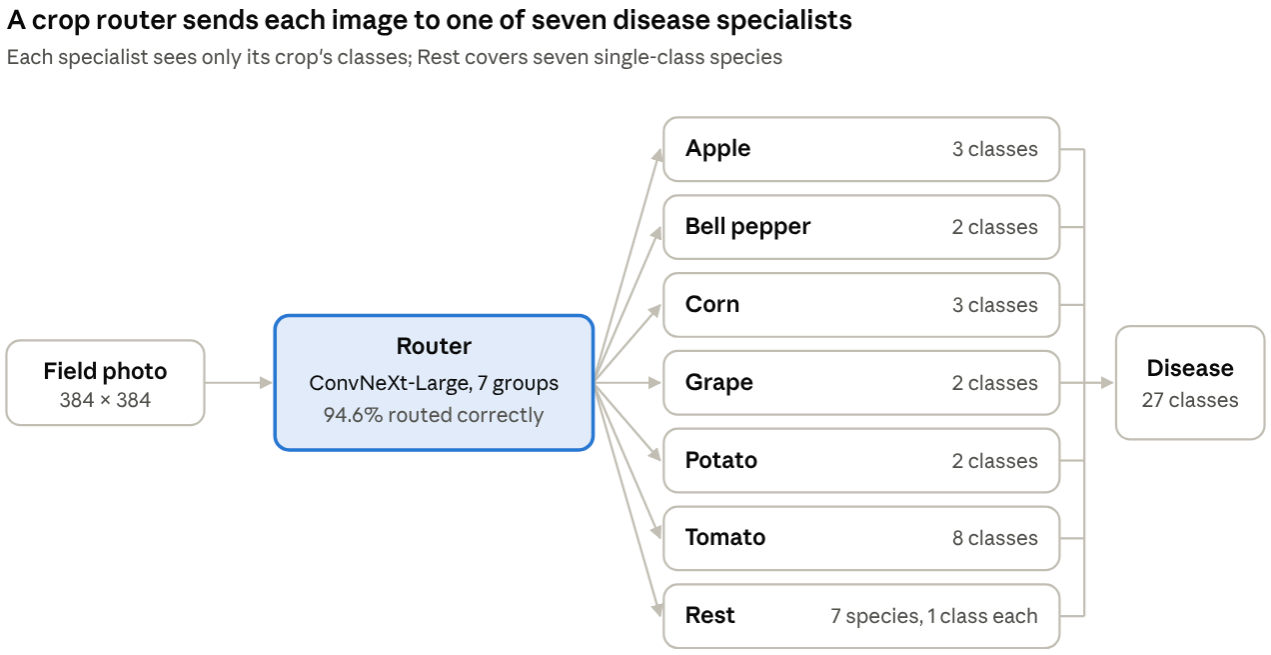
Oct 25, 2026 · @Anas Alsirafy

Overview

SpecialistRoute classifies plant diseases in field photos (PlantDoc, 27 classes) in two steps: a router identifies the crop, then a crop-specific specialist identifies the disease. On the official test set it reaches **72.80% ± 1.57%** as a single model (5 seeds) and **75.42%** as a 5-seed ensemble, under a leakage-free protocol.

- **Where:** CERERE, American University in Cairo, supervised by Prof. Dr. Maki K. Habib.
- **Goal:** a methodologically clean, publishable paper on PlantDoc.
- **State:** the v2 protocol, training of all 8 components × 5 seeds, pipeline evaluation, ensembling and the main analyses are complete. Open: one fine-tuning check, the flat baseline the paper needs, a blind test audit, and updates to older v1 documents.
- **The most useful finding so far** is methodological: evaluation shortcuts in the project's own v1 had inflated accuracy by 11.5 points (84.27% → 72.80%).

Pipeline at a glance



SpecialistRoute pipeline · 1 router, 7 specialists, 27 classes

Every component is trained separately, 5 seeds each. At test time the router picks one crop group and only that specialist predicts the disease, so a routing mistake cannot be corrected later.

The handover documents

#	Document	What it holds
1	This guide	Overview, how to navigate, status, key locations, terms
2	SpecialistRoute Handover — 2. Environment Setup	Machine, `tf_env` virtual environment, packages, weights, folder layout, first-run checks
3	SpecialistRoute Handover — 3. Data	Splits and counts, leakage removal, validation split, restored test images, CLAHE, report files
4	SpecialistRoute Handover — 4. Configs	`CONFIG_BASE`, per-component presets, evaluation settings, how to change settings safely
5	SpecialistRoute Handover — 5. Experiments	Every experiment and its status, code files and notebook sections, how to reproduce v2, output files
6	Doc	Headline numbers, per-component and per-species results, v1 → v2 breakdown, label quality, literature
7	Doc	The fine-tuning risk (check first), every gotcha with its fix, limitations to disclose
8	Doc	Prioritised next steps as checklists

How to navigate

Read this guide first. After that, the order depends on what you need:

If you need to…	Read, in order
Take over the project	1 → 7 (top risk) → 2 → 3 → 4 → 5 → 6 → 8

If you need to…	Read, in order
Run something today	2 (first-run checklist) → 5 (notebook sections) → 7 (gotchas)
Write the paper	6 → 7 (limitations) → 8 (needed for the paper) → 3 (protocol details)
Change a model or setting	4 → 7 → 5 (reproduce)
Understand a number	6, then the output file named in 5

Every document opens with its main point. Headings are written to be scanned, and tables carry the detail.

Status at a glance

Area	Status	Details in
Dataset v2 (leakage removed, official test, validation split)	Done, verified	3
8 components × 5 seeds trained	Done	5, 6
Assembled pipeline evaluated per seed	Done (72.80% ± 1.57%)	6
Seed ensemble, TTA and routing variants	Done (75.42%; no TTA, hard routing)	6
v1 → v2 breakdown (216 vs 236 test images)	Done	6
Label quality (frozen probe, 5-fold)	Scores done; human review not done	6, 8
Fine-tuning really happens?	**Unverified; check first**	7
Results workbook	Out of date (before the Potato fix)	8
Flat 27-class baseline	Not started; needed for the paper	8
Blind test audit, failure-case figure	Not started	8
v1-era documents	Still quote v1 numbers	8

Key locations

What	Where
Python environment	`/home/auc/tf_env`
Final dataset	`/home/auc/PlantDoc/Datasets/dataset/{train,val,test}`
Data-prep reports	`/home/auc/PlantDoc/Datasets/reports/`
All experiment outputs	`/home/auc/PlantDoc/Experiments/` (`v2_<Component>/`, `v2_pipeline/`, `v2_label_quality/`)
Collected results	`/home/auc/PlantDoc/Experiments/v2_all_results.xlsx`
ImageNet weights	`/home/auc/PlantDoc/Models/`
Main code	`PlantDoc_Main.ipynb`; helpers `prepare_final_dataset.py`, `verify_clahe_dataset.py`, `collect_v2_results.py`; `PlantDoc_LabelQuality.ipynb`

All settings are in section 2 of `PlantDoc_Main.ipynb`. Nothing outside the files above is needed to reproduce v2.

Terms used

Term	Meaning
v1 / v2	v1 = the earlier protocol (checkpoints chosen on test, cleaned test set); its numbers are superseded. v2 = the current protocol
Router / Classifier	ConvNeXt-Large model that assigns each image to one of 7 groups: Apple, Bell pepper, Corn, Grape, Potato, Tomato or Rest
Specialist	Per-crop disease model that receives the images the router sends it

Term	Meaning
Rest	One grouped model for the 7 species with a single class each (Blueberry, Cherry, Peach, Raspberry, Soyabean, Squash, Strawberry)
Hard routing	Each image goes only to the router's top species
Single model	One seed of every component; reported as mean ± SD over 5 seeds
Ensemble	Softmax average of the 5 seeds, for router and specialists
Restored images	20 test images removed in v1 and put back in v2 (fail the quality criteria)
216 / 236	Test set without / with the 20 restored images; 236 is the official result
Label quality	Out-of-fold score of how well an image's label agrees with a model that never saw it

Contacts

- **Previous owner:** @Anas Alsirafy (Anas Alsirafy): questions about any decision in these documents.
- **Supervisor:** Prof. Dr. Maki K. Habib, CERERE, American University in Cairo: project direction and paper decisions.